Social Network Analysis of the Canadian Disability Rights Community

Mapping Collaboration Networks and Key Actors in Disability Rights Advocacy

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2026-03-28

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1 Executive Summary

This report presents a Social Network Analysis (SNA) of collaboration patterns within the Canadian disability rights community, based on a bilingual survey (English/French). The survey, called "Enquête d'analyse des Réseaux Sociaux / Social Network Analysis Survey" enables an analysis and mapping of who collaborates with whom, identifies the most central actors and organizations, detects community clusters, and assesses overall network structure.

2 Data Cleaning and Pipeline

2.1 Source Data

The raw data was collected via a bilingual (English/French) Microsoft Forms survey distributed to members of the Canadian disability rights community. The survey instrument contained 51 columns across a single worksheet, with parallel English and French versions of each question. A total of 111 responses were recorded. The raw data file was de-identified prior to analysis.

2.2 Survey Structure and Key Fields

The survey captured several categories of information relevant to social network analysis. Respondent attributes included the respondent's connection to the disability community (e.g., person with a disability, family member/caregiver, advocate, researcher, health professional) and the disability rights issues in which they are most actively involved (free-text responses covering topics such as employment, children/youth, housing, poverty, and accessibility). Collaboration data included the names of up to three organizations or individuals with whom the respondent has partnered in the past five years (free-text), the frequency of collaboration with each named partner (regularly, occasionally, or seldom), and whether the respondent's named partners are linked to each other.

2.3 Data Cleaning Steps

The cleaning pipeline proceeded through several stages. First, the metadata row generated by Microsoft Forms was removed, and the language preference field was used to route each respondent to the appropriate English or French column for each question. Second, the bilingual fields were consolidated, with the French-language response used when the respondent selected French and the English-language response used otherwise. This produced a unified respondent table with consistent column names regardless of survey language.

Third, connection types were standardized into nine categories: Person with Disability, Advocate/CSO, Family/Caregiver, Researcher, Health, Education, Leisure, Government/Policy, and Other. Respondents whose connection type text matched multiple categories were assigned all applicable categories (semicolon-separated), with the first match designated as the primary connection type.

Fourth, disability rights issue areas were categorized from the free-text responses into 12 standardized topics: Children/Youth, Employment, Accessibility/Inclusion, Poverty/Income, Rights/Policy, Education, Housing, Health, Indigenous, Transportation, Technology, and Other. As with connection types, respondents active in multiple areas received all applicable categories.

Fifth, the free-text partner names were parsed into individual organization or person names. The parser handled several formats including comma-separated lists, semicolon-separated lists, numbered lists, and narrative descriptions. Each respondent contributed up to three partner names. Names were then normalized by converting to lowercase, collapsing whitespace, and removing punctuation. A prefix-matching algorithm was applied to merge near-duplicate entries (e.g., variant spellings or abbreviations of the same organization). This best-effort automated cleaning reduced the number of unique partner names from 175 to 166.

Sixth, collaboration frequency values were harmonized across languages, mapping both English terms (Regularly, Occasionally, Seldom) and their French equivalents (Régulièrement, Occasionnellement, Rarement) to a common three-level scale plus an Unknown category for non-standard responses.

2.4 Output Files

The pipeline produced two analysis-ready CSV files. The node table (`respondents_final.csv`) contains one row per respondent with fields for respondent ID, language, raw and categorized connection type, raw and categorized issue areas, and primary connection and issue assignments. The edge table (`edges_final.csv`) contains one row per respondent-partner tie with fields for respondent ID, raw partner name, normalized partner name, cleaned canonical partner name, raw frequency, and harmonized frequency.

3 Setup

3.1 Install Required Packages

Run this chunk once to install all dependencies. After installation, you can set `eval: false` or simply skip it on future renders.

```
install.packages(c(
  "tidyverse",      # Data wrangling and ggplot2
  "igraph",         # Core network analysis
  "tidygraph",      # Tidy interface to igraph
  "ggraph",         # Network visualisation (ggplot2-based)
  "readxl",         # Read .xlsx files
  "patchwork",      # Combine ggplot panels
  "scales",         # Axis formatting
  "RColorBrewer",   # Colour palettes
  "knitr",          # Tables in Quarto
  "kableExtra"      # Styled tables
))
```

3.2 Load Libraries

```
library(tidyverse)
library(igraph)
library(tidygraph)
library(ggraph)
library(readxl)
library(patchwork)
library(scales)
library(RColorBrewer)
library(knitr)
library(kableExtra)

theme_set(theme_minimal(base_size = 12))
set.seed(42)
```

3.3 Import Data

The analysis uses two pre-cleaned CSV files produced by Dr. Cogburn in a data-preparation pipeline. These files, `respondents_final.csv` and `edges_final.csv` are located in the data

folder, which is located in the same directory as this `.qmd` file. Paths to the data should be updated to data/filename.

```
#   Node table: one row per survey respondent
respondents <- read_csv("data/respondents_final.csv", show_col_types = FALSE) |>
  # Drop the metadata row carried over from MS Forms
filter(resp_id != "RMicrosoft Forms ")

#   Edge table: one row per respondent → partner tie
edges <- read_csv("data/edges_final.csv", show_col_types = FALSE) |>
  filter(
    !is.na(partner_clean),
    partner_clean != "nan",
    partner_clean != "None"
  )
```

```
cat(
  "Respondents:", nrow(respondents), "\n",
  "Respondents who named partners:", n_distinct(edges$respondent), "\n",
  "Unique organisations named:", n_distinct(edges$partner_clean), "\n",
  "Total collaboration ties:", nrow(edges), "\n"
)
```

```
Respondents: 111
 Respondents who named partners: 57
 Unique organisations named: 166
 Total collaboration ties: 182
```

3.4 Colour Palettes

```
# Connection-type palette (respondent node colouring)
conn_palette <- c(
  "Person with Disability" = "#E63946",
  "Advocate/CSO"           = "#457B9D",
  "Family/Caregiver"       = "#2A9D8F",
  "Researcher"             = "#6A4C93",
  "Health"                 = "#BC4749",
  "Education"              = "#F4A261",
  "Leisure"                = "#8AC926",
  "Government/Policy"      = "#264653",
  "Other"                  = "#AAAAAA",
```

```
  "Unknown"                = "#BBBBBB"
)

# Issue-area palette (bipartite squares)
issue_palette <- c(
  "Children/Youth"          = "#E63946",
  "Employment"              = "#457B9D",
  "Accessibility/Inclusion" = "#2A9D8F",
  "Poverty/Income"          = "#E9C46A",
  "Rights/Policy"           = "#264653",
  "Education"               = "#F4A261",
  "Housing"                 = "#606C38",
  "Health"                  = "#BC4749",
  "Indigenous"              = "#6A4C93",
  "Transportation"          = "#1982C4",
  "Technology"              = "#8AC926",
  "Other"                   = "#AAAAAA"
)

# Collaboration-frequency palette
freq_palette <- c(
  "Regularly"    = "#E63946",
  "Occasionally" = "#E9C46A",
  "Seldom"       = "#457B9D",
  "Unknown"      = "#BBBBBB"
)
```

4 Two-Mode Bipartite Network: Respondents × Issue Areas

This network maps respondents (circles) to the disability rights issue areas they are most actively involved in (squares). Respondents are coloured by their connection type to the disability community; issue-area nodes are sized by the number of affiliated respondents.

```
# Expand the semicolon-separated issue_category into long form
bip_edges <- respondents |>
  filter(
    !is.na(issue_category),
    issue_category != "Unknown"
  ) |>
  separate_rows(issue_category, sep = ";") |>
  mutate(issue_category = str_trim(issue_category)) |>
```

```r
  filter(issue_category != "", issue_category != "Unknown") |>
  select(resp_id, issue_category)

# Build igraph bipartite
bip_graph <- graph_from_data_frame(bip_edges, directed = FALSE)

# Mark node type: TRUE = issue-area node, FALSE = respondent node
V(bip_graph)$type <- V(bip_graph)$name %in% unique(bip_edges$issue_category)

# Attach respondent attributes
resp_lookup <- respondents |> select(resp_id, primary_connection)
V(bip_graph)$primary_connection <- resp_lookup$primary_connection[
  match(V(bip_graph)$name, resp_lookup$resp_id)
]
V(bip_graph)$primary_connection[is.na(V(bip_graph)$primary_connection)] <- "Issue Area"
```

```r
as_tbl_graph(bip_graph) |>
  mutate(
    node_type   = ifelse(type, "Issue Area", "Respondent"),
    node_degree = centrality_degree(),
    node_colour = ifelse(type, "Issue Area", primary_connection),
    node_size   = ifelse(type, node_degree * 2 + 4, node_degree + 2)
  ) |>
  ggraph(layout = "fr") +
  geom_edge_link(alpha = 0.07, colour = "#666666", width = 0.4) +
  geom_node_point(
    aes(colour = node_colour, size = node_size, shape = node_type),
    alpha = 0.75
  ) +
  geom_node_text(
    aes(label = ifelse(type, name, NA)),
    size = 3, fontface = "bold", repel = TRUE, max.overlaps = 20
  ) +
  scale_shape_manual(values = c("Issue Area" = 15, "Respondent" = 16)) +
  scale_colour_manual(
    values = c(conn_palette, "Issue Area" = "#333333"),
    name   = "Node Type"
  ) +
  scale_size_identity() +
  guides(shape = "none", size = "none") +
  labs(title = "Bipartite Network: Respondents and Disability Rights Issue Areas") +
  theme_void() +
  theme(
```

```
    legend.position = "bottom",
    plot.title = element_text(face = "bold", size = 14)
  )
```

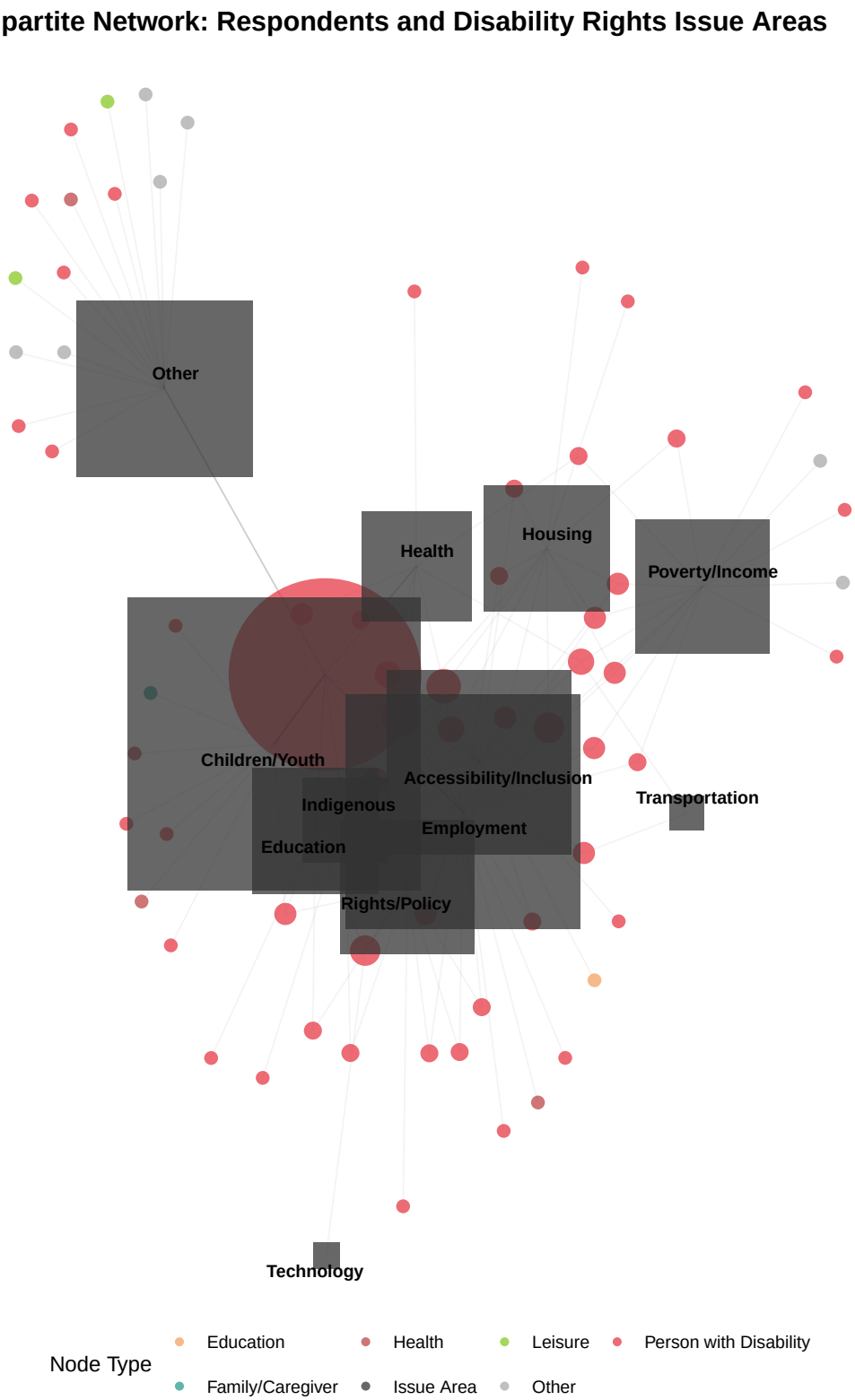

Figure 1: Figure 1. Two-mode bipartite network. Circles = respondents (coloured by connection type); squares = issue areas (sized by degree).

5 One-Mode Projected Network: Respondent Collaboration

Two respondents are linked in the projected network when they named the same organisation as a collaboration partner. Edge weight reflects the number of shared partners.

```
# Build the respondent-organisation bipartite graph
resp_org_edges <- edges |>
  select(respondent, partner_clean) |>
  distinct()

bip_ro <- graph_from_data_frame(resp_org_edges, directed = FALSE)
V(bip_ro)$type <- !(V(bip_ro)$name %in% unique(resp_org_edges$respondent))

# Project onto respondent mode
proj <- bipartite_projection(bip_ro, which = "false")

# Attach respondent attributes
V(proj)$primary_connection <- resp_lookup$primary_connection[
  match(V(proj)$name, resp_lookup$resp_id)
]
V(proj)$primary_connection[is.na(V(proj)$primary_connection)] <- "Unknown"

issue_lookup <- respondents |> select(resp_id, primary_issue)
V(proj)$primary_issue <- issue_lookup$primary_issue[
  match(V(proj)$name, issue_lookup$resp_id)
]
```

```
as_tbl_graph(proj) |>
  mutate(deg = centrality_degree()) |>
  ggraph(layout = "fr") +
  geom_edge_link(
    aes(width = weight),
    alpha = 0.25, colour = "#999999"
  ) +
  geom_node_point(
    aes(colour = primary_connection, size = deg),
    alpha = 0.8
  ) +
  geom_node_text(
    aes(label = ifelse(deg >= 2, name, NA)),
    size = 2.5, repel = TRUE, max.overlaps = 15
  ) +
  scale_edge_width(range = c(0.5, 3), name = "Shared Partners") +
```

```r
  scale_colour_manual(values = conn_palette, name = "Connection Type") +
  scale_size_continuous(range = c(3, 14), name = "Degree") +
  labs(title = "Projected Network: Respondents Linked by Shared Collaboration Partners") +
  theme_void() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(face = "bold", size = 14)
  )
```

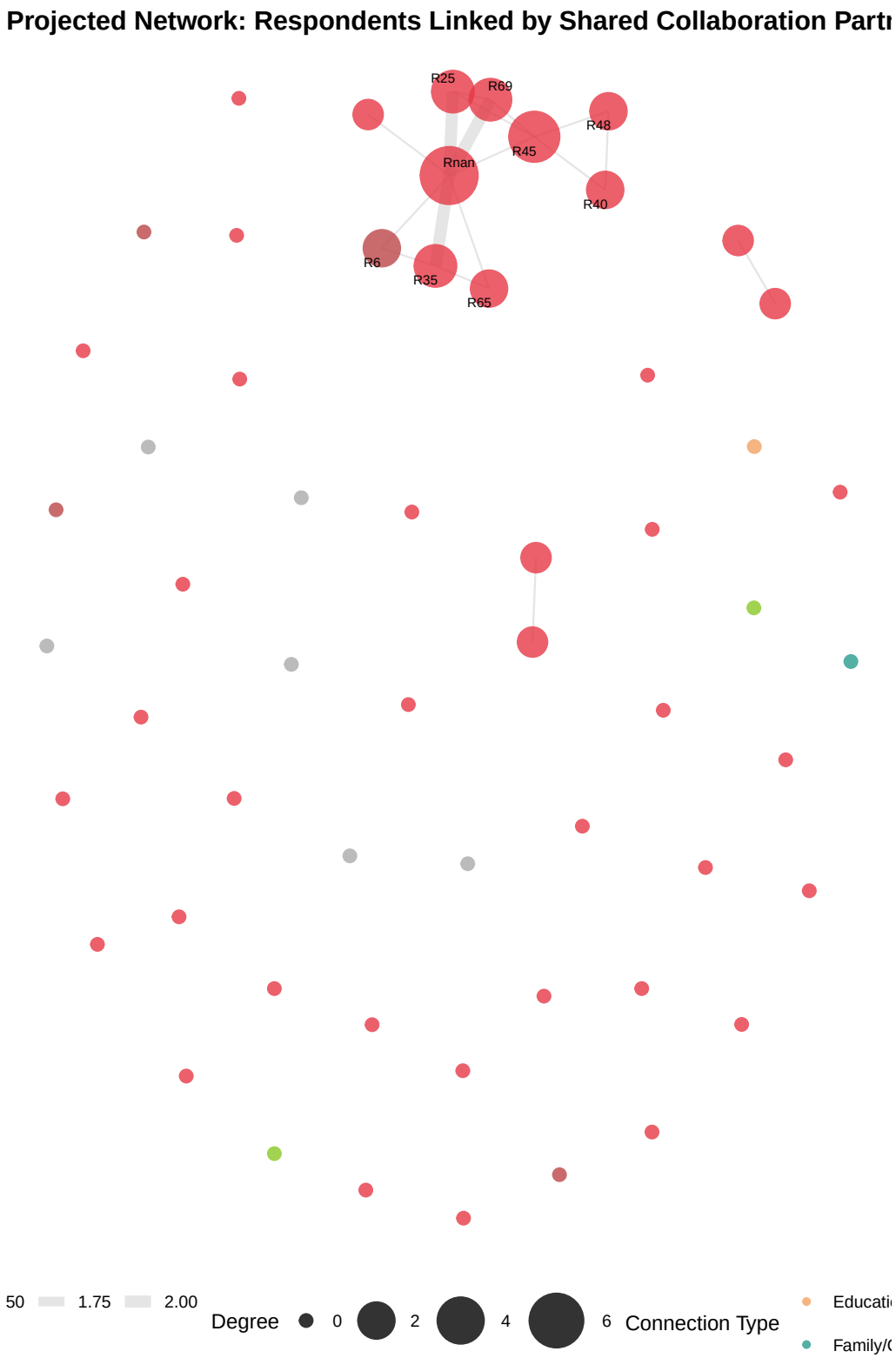

Figure 2: Figure 2. One-mode projected network. Node size = degree; colour = connection type; edge width = number of shared partners.

5.1 Network Structure Metrics

```r
metrics_tbl <- tibble(
  Metric = c(
    "Nodes (respondents with ties)",
    "Edges (shared-partner links)",
    "Density",
    "Connected components",
    "Transitivity (global clustering)",
    "Average clustering coefficient",
    "Largest component size"
  ),
  Value = c(
    vcount(proj),
    ecount(proj),
    round(graph.density(proj), 4),
    components(proj)$no,
    round(transitivity(proj, type = "global"), 4),
    round(transitivity(proj, type = "average"), 4),
    max(components(proj)$csize)
  )
)

metrics_tbl |>
  kable(align = c("l", "r"), booktabs = TRUE) |>
  kable_styling(
    bootstrap_options = c("striped", "hover"),
    latex_options = c("hold_position"),
    full_width = FALSE
  )
```

Metric	Value
Nodes (respondents with ties)	57.0000
Edges (shared-partner links)	17.0000
Density	0.0107
Connected components	46.0000
Transitivity (global clustering)	0.4773
Average clustering coefficient	0.8116
Largest component size	10.0000

6 Centrality Analysis

Centrality measures identify the most influential actors. **Degree centrality** counts direct connections; **betweenness centrality** identifies actors who bridge otherwise disconnected parts of the network.

6.1 Organisation Degree Centrality

```
org_degree <- edges |>
  count(partner_clean, sort = TRUE, name = "degree") |>
  head(15)

ggplot(org_degree, aes(x = reorder(partner_clean, degree), y = degree)) +
  geom_col(fill = "#457B9D", alpha = 0.85, width = 0.7) +
  geom_text(aes(label = degree), hjust = -0.3, size = 3.5) +
  coord_flip() +
  labs(
    x = NULL,
    y = "Degree (respondents naming this organisation)",
    title = "Organisation Degree Centrality"
  ) +
  theme(plot.title = element_text(face = "bold"))
```

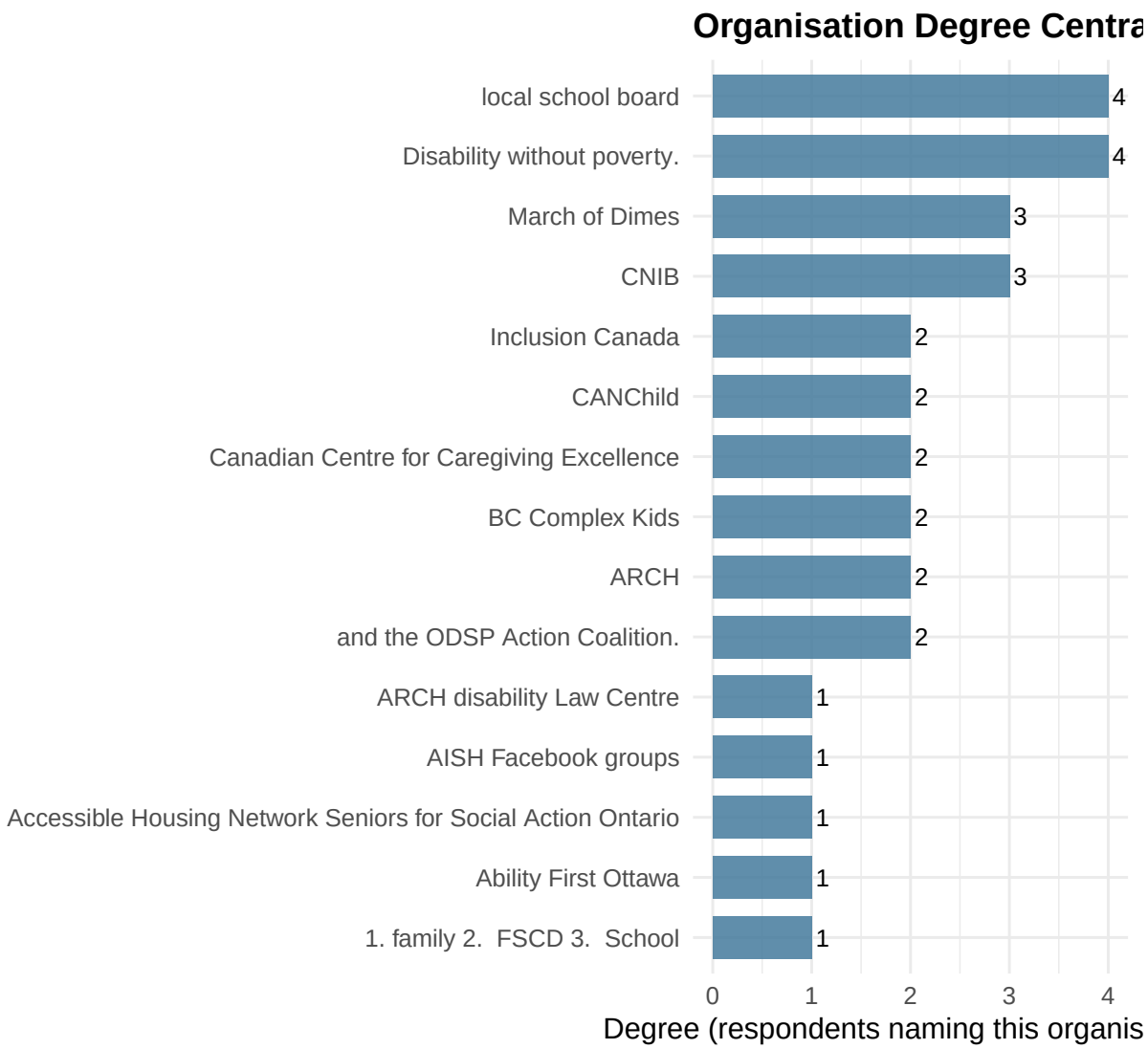

Figure 3: Figure 3a. The 15 most frequently named collaboration partners.

6.2 Respondent Centrality (Projected Network)

```
# Compute centrality measures
cent_df <- tibble(
  name               = V(proj)$name,
  primary_connection = V(proj)$primary_connection,
  degree             = degree(proj),
  betweenness        = betweenness(proj, weights = E(proj)$weight)
```

```r
) |>
  arrange(desc(degree))

# Degree panel
p_deg <- cent_df |>
  head(12) |>
  mutate(label = paste0(name, " (", str_sub(primary_connection, 1, 12), ")")) |>
  ggplot(aes(x = reorder(label, degree), y = degree, fill = primary_connection)) +
  geom_col(alpha = 0.85, width = 0.7) +
  coord_flip() +
  scale_fill_manual(values = conn_palette) +
  labs(x = NULL, y = "Degree", title = "Respondent Degree Centrality") +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold")
  )

# Betweenness panel
p_betw <- cent_df |>
  arrange(desc(betweenness)) |>
  head(12) |>
  mutate(label = paste0(name, " (", str_sub(primary_connection, 1, 12), ")")) |>
  ggplot(aes(x = reorder(label, betweenness), y = betweenness, fill = primary_connection)) +
  geom_col(alpha = 0.85, width = 0.7) +
  coord_flip() +
  scale_fill_manual(values = conn_palette) +
  labs(x = NULL, y = "Betweenness", title = "Respondent Betweenness Centrality") +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold")
  )

p_deg + p_betw +
  plot_annotation(
    title = "Centrality Analysis: Key Actors in the Projected Network",
    theme = theme(plot.title = element_text(face = "bold", size = 14))
  )
```

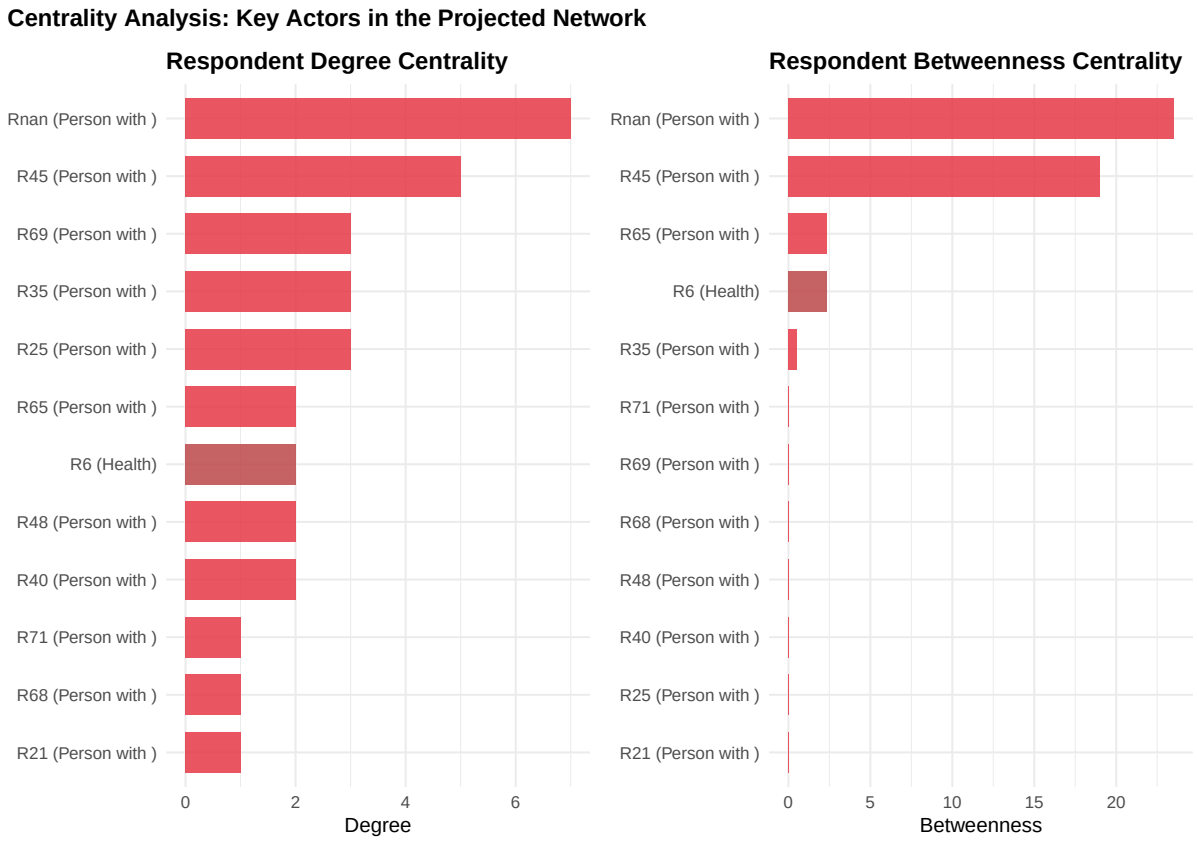

Figure 4: Figure 3b. Respondent degree and betweenness centrality in the projected network.

7 Community Detection

The Louvain algorithm partitions the network into communities that maximise modularity — groups of respondents who are more densely connected to each other than to the rest of the network.

```
comms <- cluster_louvain(proj, weights = E(proj)$weight)

# Summary stats
n_comms    <- length(unique(membership(comms)))
mod_score  <- round(modularity(comms), 4)
comm_sizes <- sort(table(membership(comms)), decreasing = TRUE)

cat(
  "Communities detected:", n_comms, "\n",
```

```r
  "Modularity:", mod_score, "\n",
  "Sizes:", paste(comm_sizes, collapse = ", "), "\n"
)
```

```
Communities detected: 47
 Modularity: 0.3387
 Sizes: 7, 3, 2, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
```

```r
as_tbl_graph(proj) |>
  mutate(
    community = as.factor(membership(comms)),
    deg       = centrality_degree()
  ) |>
  ggraph(layout = "fr") +
  geom_edge_link(alpha = 0.15, colour = "#BBBBBB", width = 0.5) +
  geom_node_point(
    aes(colour = community, size = deg),
    alpha = 0.85
  ) +
  geom_node_text(
    aes(label = ifelse(deg >= 2, name, NA)),
    size = 2.5, repel = TRUE, max.overlaps = 15
  ) +
  scale_size_continuous(range = c(3, 14), name = "Degree") +
  labs(
    title   = paste0("Community Detection (Louvain): ", n_comms, " Communities"),
    subtitle = paste0("Modularity = ", mod_score),
    colour  = "Community"
  ) +
  theme_void() +
  theme(
    legend.position = "bottom",
    plot.title    = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11, colour = "grey40")
  )
```

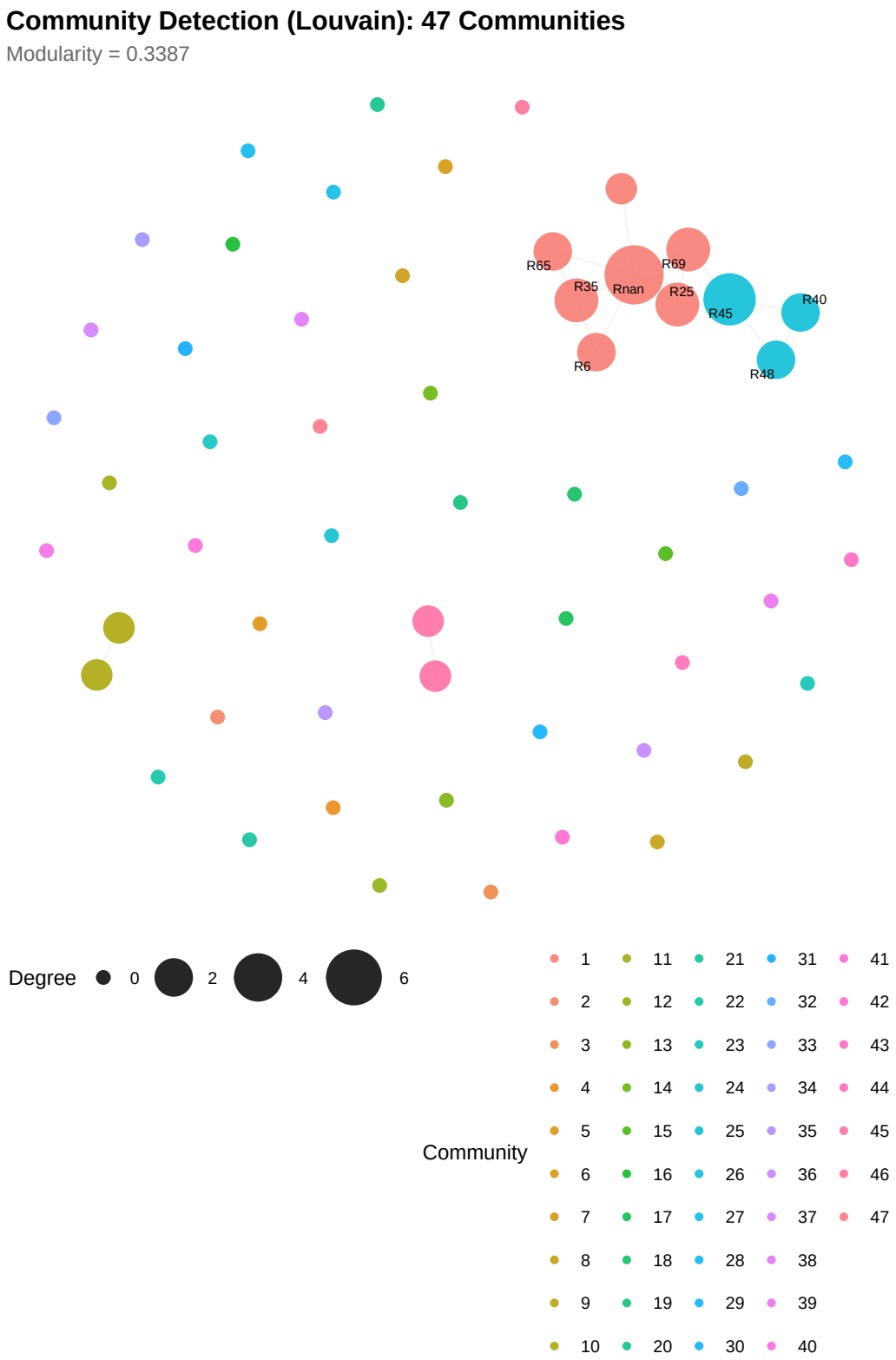

Figure 5: Figure 4. Louvain communities. Colour = community membership; size = degree.

7.1 Community Composition

```r
# Build composition summary for multi-member communities
comm_membership <- tibble(
  name               = V(proj)$name,
  primary_connection = V(proj)$primary_connection,
  primary_issue      = V(proj)$primary_issue,
  community          = membership(comms)
)

comm_summary <- comm_membership |>
  group_by(community) |>
  summarise(
    Size = n(),
    `Connection Types` = paste(
      names(sort(table(primary_connection), decreasing = TRUE)[1:min(3, n())]),
      collapse = ", "
    ),
    `Issue Areas` = paste(
      names(sort(table(primary_issue), decreasing = TRUE)[1:min(3, n())]),
      collapse = ", "
    ),
    .groups = "drop"
  ) |>
  filter(Size >= 2) |>
  arrange(desc(Size)) |>
  mutate(Community = paste0("Community ", row_number())) |>
  select(Community, Size, `Connection Types`, `Issue Areas`)

comm_summary |>
  kable(booktabs = TRUE) |>
  kable_styling(
    bootstrap_options = c("striped", "hover"),
    latex_options = c("hold_position", "scale_down"),
    full_width = FALSE
  )
```

Community	Size	Connection Types	Issue Areas
Community 1	7	Person with Disability, Health, NA	Children/Youth, Employment, Housing
Community 2	3	Person with Disability, NA, NA	Employment, Housing, Other
Community 3	2	Person with Disability, NA	Employment, Poverty/Income

Community 4	2	Person with Disability, NA	Children/Youth, NA

8 Degree Distributions and Collaboration Patterns

```
# Organisation degree distribution
p1 <- edges |>
  count(partner_clean, name = "degree") |>
  ggplot(aes(x = degree)) +
  geom_histogram(
    binwidth = 1, fill = "#457B9D", alpha = 0.8,
    colour = "white", boundary = 0.5
  ) +
  geom_vline(
    aes(xintercept = mean(degree)),
    linetype = "dashed", colour = "#E63946"
  ) +
  labs(
    x = "Degree", y = "Count",
    title = "Organisation Degree Distribution"
  ) +
  theme(plot.title = element_text(face = "bold", size = 10))

# Respondent degree distribution (projected network)
resp_deg_df <- tibble(degree = degree(proj))

p2 <- ggplot(resp_deg_df, aes(x = degree)) +
  geom_histogram(
    binwidth = 1, fill = "#2A9D8F", alpha = 0.8,
    colour = "white", boundary = -0.5
  ) +
  geom_vline(
    aes(xintercept = mean(degree)),
    linetype = "dashed", colour = "#E63946"
  ) +
  labs(
    x = "Degree", y = "Count",
    title = "Respondent Degree Distribution\n(Projected Network)"
  ) +
  theme(plot.title = element_text(face = "bold", size = 10))
```

```r
# Collaboration frequency
p3 <- edges |>
  filter(freq_clean != "Unknown") |>
  count(freq_clean) |>
  ggplot(aes(x = freq_clean, y = n, fill = freq_clean)) +
  geom_col(alpha = 0.85, width = 0.7) +
  geom_text(aes(label = n), vjust = -0.4, fontface = "bold", size = 3.5) +
  scale_fill_manual(values = freq_palette) +
  labs(x = NULL, y = "Count", title = "Collaboration Frequency") +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold", size = 10)
  )

p1 + p2 + p3 +
  plot_annotation(
    title = "Network Structure: Degree Distributions and Collaboration Patterns",
    theme = theme(plot.title = element_text(face = "bold", size = 14))
  )
```

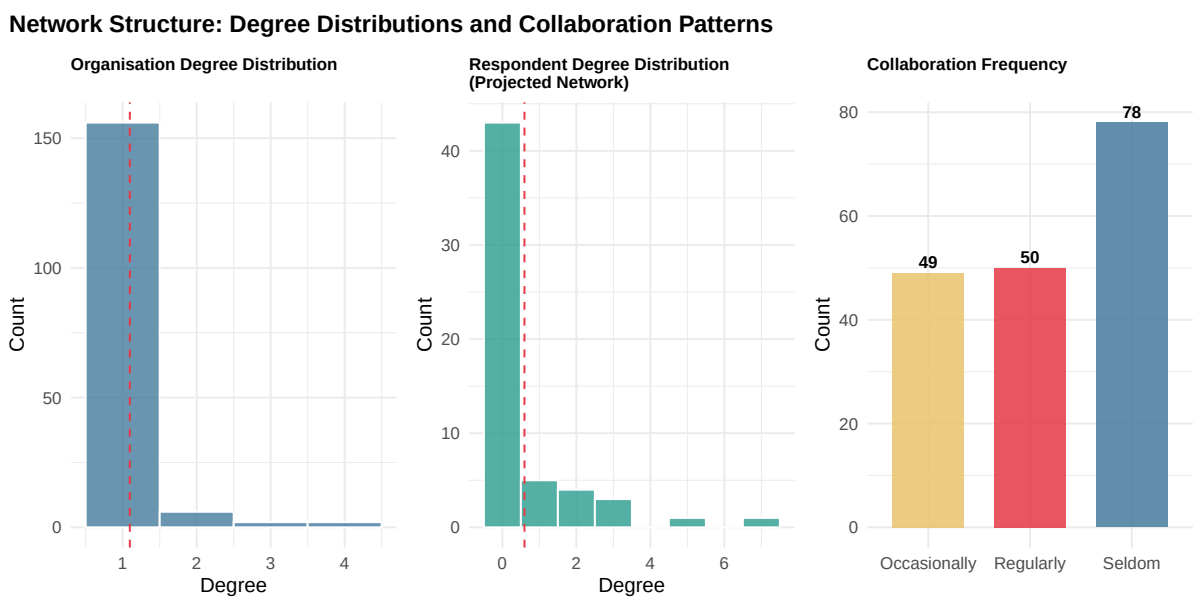

Figure 6: Figure 5. Left: Organisation degree distribution. Centre: Respondent degree distribution (projected). Right: Collaboration frequency.

9 Discussion and Implications

The Canadian disability rights community operates as a **highly decentralized network**. Collaboration is predominantly local and issue-specific rather than nationally integrated. A small number of national organizations serve as critical connectors whose capacity to convene and bridge across issue silos is disproportionately important to overall network cohesion.

9.1 Fragmentation as a structural feature

The Canadian disability rights community, as captured by this survey, operates as a highly decentralised network. Collaboration is predominantly local and issue-specific rather than nationally integrated. This is not necessarily a weakness — it may reflect the diversity of disability issues and the importance of place-based advocacy — but it does mean that information, resources, and strategic coordination may not flow easily across the network.

9.2 The role of national organisations

A small number of national organisations (Disability Without Poverty, CNIB, March of Dimes, Inclusion Canada) serve as critical connectors. These organisations' capacity to convene, share information, and bridge across issue silos is disproportionately important to overall network cohesion. Strengthening these hub organisations — or creating new ones — could substantially improve coordination across the community.

9.3 Bridging actors

Respondents with high betweenness centrality are structurally important: they connect otherwise isolated parts of the network. Supporting and amplifying these bridging actors — many of whom are persons with disabilities themselves — could enhance cross-pollination of ideas and strategies.

9.4 Limitations

This analysis is based on a convenience sample of 111 respondents. The free-text partner naming format means some organisations may be inconsistently recorded. The projected one-mode network relies on the assumption that shared organisational partnerships indicate a meaningful indirect connection between respondents. Finally, the survey captures reported collaboration, which may differ from actual interaction patterns.

Report generated from the Enquête d'analyse des Réseaux Sociaux / Social Network Analysis Survey. Analysis performed in R using igraph, tidygraph, and ggraph. All respondent identifiers are de-identified survey IDs.